

Battery Tester

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Here is a simple circuit to test a wide range of batteries quickly. The circuit does not have a power supply of its own and uses the power from the battery under test for its operation. It can test the batteries in 3V-15V range, which makes it a very useful tool for a field person working on battery-operated systems.

As the system works with power derived from the battery under test, it saves you money that you would otherwise spend on batteries if you were using other similar battery-operated systems.

Circuit and working

Fig. 1 shows the circuit of battery tester. MOSFET BS170/2N3819 (T1) provides a constant current drive for LED1 and biasing to transistor T2. In this way, LED1 illuminates at a constant intensity, independent of battery voltage from 3V-15V, and T2 applies a constant current load of about 120mA to the battery when S1 is pressed.

IC1 is wired to generate a square wave of about 3kHz. IC2 acts as an inverter triggered by the output of

IC1. The outputs of IC1 and IC2 are opposite in phase to each other. Diodes D2 through D5 and capacitors C4 through C7 are used for voltage multiplication. Electrolytic capacitor C8 is charged by this multiplied voltage. Resistor R3 and potmeter VR1 form a voltage divider for biasing the base of transistor T3.

When S1 is open, a very light load gets applied to the battery under test and the base of transistor T3 gets biased through VR1 and so keeps LED2 in the off state. Closing S1 applies a 120mA load to the battery under test. If the battery is not fully charged, its output voltage starts reducing quickly. When this voltage falls below 0.6V of the battery's nominal voltage, transistor T3's emitter becomes more negative than its base and LED2 illuminates. This state lasts only for a few seconds, depending on the discharge



rate of capacitor C8. If the battery under test is in a good charged state, its output voltage will not fall under the 120mA loading current and LED2 will stay in its off state.

Construction and testing

An actual-size, single-side PCB for the battery tester is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose

PARTS LIST

Semiconductors:

- IC1, IC2 - NE555 timer
- T1 - MOSFET BS170
- T2, T3 - BC337 npn transistor
- LED1, LED2 - 5mm LED
- D1-D5 - 1N4148 diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 2.2-kilo-ohm
- R2 - 3.3-ohm
- R3 - 500-kilo-ohm
- R4 - 4.7-kilo-ohm
- R5 - 33-kilo-ohm
- R6, R7 - 100-kilo-ohm
- VR1 - 500-kilo-ohm preset

Capacitors:

- C1, C2, C9, C10 - 0.01 μ F ceramic disk
- C3-C7 - 0.1 μ F ceramic disk
- C8 - 220 μ F, 25V electrolytic

Miscellaneous:

- S1 - Push-to-switch
- *BUT - 2-pin terminal block

Test Points

Test point	Details
TP0	GND
TP1	Battery voltage
TP2	3kHz
TP3	12V-25V depending on the battery

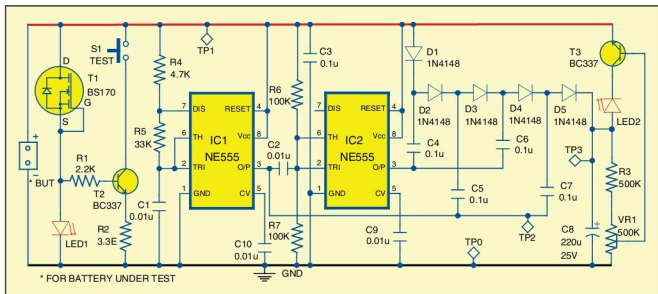


Fig. 1: Circuit diagram of the battery tester